

R PROGRAMMING – ASSESSMENT TOOL 1

1. Variables and Data Types Analysis (BL4 – Analyze)

Design pseudocode to simulate basic variable handling in R programming:

- Declare variables of different data types (numeric, character, logical)
- Perform arithmetic and logical operations
- Display the results

Pseudo Code

Start

Declare num as numeric

Declare name as character

Declare flag as logical

Assign values to num, name and flag

Add 10 to num

Perform logical AND operation

Display num

Display name

Display flag

Stop

R Program Code

```
num <- 20
```

```
name <- "Sahil"
```

```
flag <- TRUE
num <- num + 10
flag <- flag & TRUE
print(num)
print(name)
print(flag)
```

Output

```
> num <- 20
> name <- "Sahil"
> flag <- TRUE
> num <- num + 10
> flag <- flag & TRUE
>
> print(num)
[1] 30
> print(name)
[1] "Sahil"
> print(flag)
[1] TRUE
> |
```

2. Data Structures Implementation (BL4 – Analyze)

Write pseudocode to demonstrate advanced data structures in R:

- Create and initialize a vector, list, and matrix
- Perform operations such as indexing, updating, and aggregation
- Compare the structures based on usage

Pseudo Code

Start

Create a vector

Create a list

Create a matrix

Display second element of vector

Update first element of vector

Find sum of vector

Display updated vector

Display list

Display matrix

Stop

R Program Code

```
vec <- c(10,20,30,40)
```

```
lst <- list("R",100,TRUE)
```

```
mat <- matrix(c(1,2,3,4), nrow=2)
```

```
print(vec[2])
```

```
vec[1] <- 50
```

```
print(sum(vec))
```

```
print(vec)
```

```
print(lst)
```

```
print(mat)
```

Output

```
> vec <- c(10,20,30,40)
> lst <- list("R",100,TRUE)
> mat <- matrix(c(1,2,3,4), nrow=2)
> print(vec[2])
[1] 20
> vec[1] <- 50
> print(sum(vec))
[1] 140
> print(vec)
[1] 50 20 30 40
> print(lst)
[[1]]
[1] "R"

[[2]]
[1] 100

[[3]]
[1] TRUE

> print(mat)
      [,1] [,2]
[1,]    1    3
[2,]    2    4
> |
```

3. Matrix and Array Operations (BL4 – Analyze)

Design pseudocode to perform operations on matrices and arrays:

- Create a matrix and a 3D array
- Perform addition, multiplication (matrix), and element-wise operations
- Display the results and analyze differences

Pseudo Code

Start

Create two matrices

Create a 3D array

Add the matrices

Multiply the matrices

Perform element-wise multiplication

Display matrix addition

Display matrix multiplication

Display element-wise multiplication

Display array

Stop

R Program Code

```
m1 <- matrix(c(1,2,3,4), nrow=2)
```

```
m2 <- matrix(c(5,6,7,8), nrow=2)
```

```
arr <- array(1:8, dim=c(2,2,2))
```

```
add <- m1 + m2
```

```
mul <- m1 %*% m2
```

```
ele <- m1 * m2
```

```
print(add)
```

```
print(mul)
```

```
print(ele)
```

```
print(arr)
```

Output

```
> m1 <- matrix(c(1,2,3,4), nrow=2)
> m2 <- matrix(c(5,6,7,8), nrow=2)
> arr <- array(1:8, dim=c(2,2,2))
> add <- m1 + m2
> mul <- m1 %*% m2
> ele <- m1 * m2
> print(add)
      [,1] [,2]
[1,]    6   10
[2,]    8   12
> print(mul)
      [,1] [,2]
[1,]   23   31
[2,]   34   46
> print(ele)
      [,1] [,2]
[1,]    5   21
[2,]   12   32
> print(arr)
, , 1
      [,1] [,2]
[1,]    1    3
[2,]    2    4

, , 2
      [,1] [,2]
[1,]    5    7
[2,]    6    8

> |
```

4. Data Frame Construction and Manipulation (BL4 – Analyze))

A dataset needs to be organized efficiently.

Write pseudocode to:

- Create a data frame with multiple columns (numeric and character)
- Add, delete, and modify columns
- Perform basic summary operations

Pseudo Code

Start

Create a data frame

Add a new column

Modify marks column

Delete age column

Display data frame

Display summary

Stop

R Program Code

```
df <- data.frame(  
  Name=c("A","B","C"),  
  Marks=c(80,75,90)  
)  
df$Age <- c(18,19,20)  
df$Marks <- c(85,78,95)  
df$Age <- NULL  
print(df)  
summary(df)
```

Output

```
> df <- data.frame(
+   Name=c("A","B","C"),
+   Marks=c(80,75,90)
+ )
> df$Age <- c(18,19,20)
> df$Marks <- c(85,78,95)
> df$Age <- NULL
> print(df)
  Name Marks
1    A    85
2    B    78
3    C    95
> summary(df)
      Name      Marks
Length   :3   Min.   :78.0
N.unique :3   1st Qu.:81.5
N.blank  :0   Median :85.0
Min.nchar:1   Mean    :86.0
Max.nchar:1   3rd Qu.:90.0
              Max.    :95.0
> |
```

5. Functions and Logical Operations (BL4 – Analyze)

Design pseudocode to implement functions and logical vector operations:

- Create a user-defined function with input parameters
- Perform operations on logical vectors
- Return and display computed results

Pseudo Code

Start

Create a function to add two numbers

Pass two numbers to the function

Store the result

Create a logical vector

Perform logical AND operation

Display result

Display logical vector

Stop

R Program Code

```
add <- function(a,b)
{
  a+b
}
result <- add(15,20)
logic <- c(TRUE,FALSE,TRUE)
ans <- logic & TRUE
print(result)
print(ans)
```

Output

```
> add <- function(a,b)
+ {
+   a+b
+ }
> result <- add(15,20)
> logic <- c(TRUE,FALSE,TRUE)
> ans <- logic & TRUE
> print(result)
[1] 35
> print(ans)
[1] TRUE FALSE TRUE
> |
```